

Maths Week 1: Equations

Written application (20 to 30 minutes). Pen or pencil, no calculator. Full working in every box.

The one habit this week: working, then a check

One step per line. Say what you did to **both sides** in brackets. Finish with a **check**: put your x back into the original equation and show that both sides match.

Worked example $5x - 2 = 18$

$5x = 20$ (add 2 to both sides)

$x = 4$ (divide both sides by 5)

Check: $5(4) - 2 = 18$ ✓

Part A: Solve and check (15 minutes)

Q1 $5x + 3 = 38$

Check: _____

Q2 $4(x - 2) = 28$

Check: _____

Q3 $6x - 5 = 2x + 19$

Check: _____

Q4 $3(2x + 1) = 5x + 11$

Check: _____

Part B: Forming equations (10 minutes)

Q5. Ali thinks of a number. He multiplies it by 3, then subtracts 4. The answer is 20. **Form an equation and solve it** to find Ali's number. (3 marks)

Check: _____

Q6. A rectangle has length $(x + 5)$ cm and width x cm. Its perimeter is 38 cm.

(a) Show that $4x + 10 = 38$. (b) Solve it to find x . (c) Work out the area of the rectangle. (5 marks)

Check: _____

Self-check (5 minutes)

- ☐ Every equation has one step per line, with what I did to **both sides** written beside it.
- ☐ Every answer has a check line where I substituted into the **original** equation.
- ☐ In each check, both sides really did match. If one did not, I found and fixed the mistake.
- ☐ In Q2 and Q4, I expanded the bracket first and multiplied **every** term inside.
- ☐ In Q5 and Q6, I wrote the equation first before solving, and I answered the actual question (Ali's number, the area).

Oral check with a parent (5 minutes)

Parent: pick one equation and ask: “**Why did you do that to both sides?**” Then: “**How do you know your answer is right?**” She should say the check, not “I just know”. Parent key below.

Answered without reading it ☐ Needed a prompt ☐ Could not answer ☐

Parent key

Q1 $5x = 35$, $x = 7$. Check $5(7) + 3 = 38$. **Q2** $4x - 8 = 28$, $4x = 36$, $x = 9$. Check $4(7) = 28$.

Q3 $4x - 5 = 19$, $4x = 24$, $x = 6$. Check $6(6) - 5 = 31$ and $2(6) + 19 = 31$.

Q4 $6x + 3 = 5x + 11$, $x + 3 = 11$, $x = 8$. Check $3(17) = 51$ and $5(8) + 11 = 51$.

Q5 $3x - 4 = 20$, $3x = 24$, $x = 8$. Check $3(8) - 4 = 20$.

Q6 (a) $2(x+5) + 2x = 38$, so $4x + 10 = 38$. (b) $4x = 28$, $x = 7$. (c) Length 12 cm, width 7 cm, area 84 cm^2 . Check perimeter $2(12) + 2(7) = 38$.

Weekly review (for J, 5 minutes)

Timings

Screen drill: _____ min Written: _____ min Self-check: _____ min

What was weak? (e.g. vague evidence, skipped working, skipped the check)

What worked?

Change for Week 2 (feeds the next build; compare with this week's Sparx codes once known):